Ans: C

7 The engine of a boat supplies a constant power of 110 kW to propel the boat forward. The boat attains a maximum speed of 21.0 m s^{-1}.

If the magnitude of the resistive force acting on the boat is proportional to the square of the boat's speed, what is the resultant force acting on the boat when it is moving at the instant when its speed is 15.0 m s^{-1}?

A 2.7 kN **B** 3.6 kN **C** 4.7 kN **D** 7.3 kN

Ans: C

Drag @ 21.0 m s^{-1}: $f_D = \dfrac{110}{21} = 5.24$ kN

Drag @ 15.0 m s^{-1}: $\dfrac{f_D'}{f_D} = \left(\dfrac{15.0}{21.0}\right)^2 \Rightarrow f_D' = 2.67$ kN

Force of engine @ 15.0 m s^{-1}: $F_{engine} = \dfrac{110}{15} = 7.33$ kN

Hence resultant force = 7.33 – 2.67 = 4.66 kN

8 A crane is used to raise a weight of 200 N at a constant speed through a vertical height of